

Multiplicity Runtime — A Mathematical Note

(IFMD v0.1)

Purpose. Provide a compact, formal specification and basic guarantees for a prime-graded, contractive runtime used to update a state in a Banach space under explicit safety, structure, and governance constraints.

0. Preliminaries & Notation

- Let $(V, \|\cdot\|)$ be a real (or complex) Banach space; $\mathcal{L}(V)$ denotes bounded linear operators on V .
- Fix an index set $\mathcal{P} \subseteq \{\text{primes}\}$. For each $p \in \mathcal{P}$ there is a channel operator $B_p \in \mathcal{L}(V)$ with a known bound $b_p \geq \|B_p\|$.
- Time is discrete: $t = 0, 1, 2, \dots$. The runtime maintains a state $T_t \in V$ and receives a forcing $F \in V$ (assumed time-invariant in the base model; see §4.3 for time-varying variants).
- At time t a finite active set $\mathcal{P}_t \subset \mathcal{P}$ and a weight vector $w_t = (w_{p,t})_{p \in \mathcal{P}_t} \in \mathbb{R}^{\mathcal{P}_t}$ are chosen.
- **Composite operator:** $K_t := \sum_{p \in \mathcal{P}_t} w_{p,t} B_p \in \mathcal{L}(V)$, with the norm bound

$$\|K_t\| \leq \sum_{p \in \mathcal{P}_t} b_p |w_{p,t}|.$$

- **State update:**

$$T_{t+1} = F + K_t T_t. \tag{U}$$

Design remark. The runtime can compute **proposal weights** $\tilde{w}_t$ from an upstream learner and then impose safety/structure via projection and invariants (§1–§3). The mathematics below concerns the *accepted* weights w_t .

1. ACE Safety Set & Budgeted Projection

Let a measurable scalar magnitude $X_t \in [0, 1)$ be supplied (e.g., a norm bound, telemetry magnitude, or certified upper bound on a nominal sub-term). Let $\varepsilon_t \in (0, 1)$ be a **gap parameter**. Define the **ACE budget radius**

$$R_t := 1 - \varepsilon_t - X_t.$$

The **ACE feasible set** at time t is the weighted ℓ_1 -ball

$$\mathcal{S}_t := \left\{ w \in \mathbb{R}^{\mathcal{P}_t} : \sum_{p \in \mathcal{P}_t} b_p |w_p| \leq R_t \right\}. \tag{ACE}$$

Given any proposal $\tilde{w}_t \in \mathbb{R}^{\mathcal{P}_t}$, the accepted weights are the **weighted projection**

$$w_t = \arg \min_{w \in \mathcal{S}_t} \frac{1}{2} \|w - \tilde{w}_t\|_2^2. \quad (\text{Proj})$$

1.1 KKT certificate (implementation-ready)

There exists $\lambda_t \geq 0$ and subgradients $\xi_{p,t} \in \partial |w_{p,t}|$ such that

$$w_t - \tilde{w}_t + \lambda_t b_p \xi_{p,t} = 0 \quad (p \in \mathcal{P}_t), \quad \lambda_t \left(\sum_p b_p |w_{p,t}| - R_t \right) = 0, \quad \sum_p b_p |w_{p,t}| \leq R_t.$$

A standard **soft-threshold with weights** algorithm recovers w_t in $O(|\mathcal{P}_t| \log |\mathcal{P}_t|)$ time. Residuals from KKT give a numerical certificate.

1.2 Immediate contraction lemma

If $w_t \in \mathcal{S}_t$, then $\|K_t\| \leq 1 - \varepsilon_t$. Hence the step map $\Phi_t(x) := F + K_t x$ is a contraction with constant $\kappa_t := \|K_t\| \leq 1 - \varepsilon_t < 1$.

2. Baseline Guarantees (Time-Invariant Case)

Assume **time-invariant** weights ($w_t \equiv w$) so that $K_t \equiv K$ and $\|K\| \leq 1 - \varepsilon$ for some $\varepsilon \in (0, 1)$.

Theorem 2.1 (Banach fixed point). There is a unique equilibrium

$$T_* = (I - K)^{-1} F = \sum_{j \geq 0} K^j F,$$

and for any $T_0 \in V$, the trajectory of (U) satisfies

$$\|T_t - T_*\| \leq (1 - \varepsilon)^t \|T_0 - T_*\|.$$

Proof. Standard Banach contraction and Neumann series arguments. $\square$

3. Time-Varying Contraction

We now allow K_t to vary with t while maintaining contraction.

3.1 Uniform contraction

Assume **uniform gap**: $\|K_t\| \leq \kappa < 1$ for all t .

Lemma 3.1 (distance decay). For any two trajectories of (U) with initial conditions $T_0, \bar{T}_0$,

$$\|T_t - \bar{T}_t\| \leq \left(\prod_{s=0}^{t-1} \kappa \right) \|T_0 - \bar{T}_0\| \leq \kappa^t \|T_0 - \bar{T}_0\|.$$

Theorem 3.2 (Cauchy & limit). If $\|K_t\| \leq \kappa < 1$, the sequence (T_t) is Cauchy and converges to a limit $T_\infty \in V$. If additionally $K_t \rightarrow K_\infty$ in operator norm, then T_∞ is the unique solution of $T = F + K_\infty T$, i.e. $T_\infty = (I - K_\infty)^{-1}F$.

Sketch. Compose contractive maps and use completeness of V . Continuity of the fixed point under operator-norm convergence gives the last claim. $\square$

3.2 Windowed average contraction (WAC)

Fix an integer window $m \geq 1$. Suppose there exists $\rho \in (0, 1)$ such that for all t

$$\|K_{t+m-1} \cdots K_t\| \leq \rho. \quad (\text{WAC})$$

Theorem 3.3 (rate under WAC). Under (WAC), for any T_0 ,

$$\|T_t - T_{t-m}\| \leq C \rho^{\lfloor t/m \rfloor} \quad (t \geq m),$$

for some constant C depending on $\|T_0\|, \|F\|, m$. In particular, (T_t) converges and the convergence is at least geometric at rate $\rho^{1/m}$.

Idea. Consider the m -step map and apply Theorem 2.1 to the induced dynamics on blocks. $\square$

3.3 Drift with sparse expansions (DSE)

Allow occasional steps with $\|K_t\| > 1$ provided that within every macro-window of length M the product is contractive: there exist $M \in \mathbb{N}, \bar{\rho} \in (0, 1)$ s.t.

$$\max_t \|K_{t+M-1} \cdots K_t\| \leq \bar{\rho}. \quad (\text{DSE})$$

Then the conclusions of Theorem 3.3 hold with $m = M, \rho = \bar{\rho}$.

4. Moonshine Governor (Scalar Intensity Control)

Let $M(\cdot, p) : V \rightarrow \mathbb{R}_{\geq 0}$ be channel estimators with Lipschitz constants L_p . Fix $\alpha_M > 1$ and a guard $\delta_{\min} > 0$. Define the **governor**

$$\Xi_t := \left(\sum_{p \in \mathcal{P}_t} M(T_t, p) p^{-\alpha_M} + \delta_{\min} \right)^{-1}.$$

This scalar can be used to scale proposals, e.g. $\hat{w}_{p,t} \mapsto \Xi_t \hat{w}_{p,t}$, before the ACE projection.

Proposition 4.1 (small-gain feasibility). If a proposal $\hat{w}_t$ satisfies $\sum_p b_p |\hat{w}_{p,t}| \leq S_t$ and we choose $\Xi_t \leq R_t/S_t$, then the scaled vector $\Xi_t \hat{w}_t$ lies in $\mathcal{S}_t$. Consequently the subsequent projection is a no-op and the ACE margin equals $R_t - \sum_p b_p |\Xi_t \hat{w}_{p,t}| \geq 0$.

Remark (summability). If $\mathcal{P}_t$ may be infinite, $\alpha_M > 1$ ensures the Dirichlet weight $\sum_p p^{-\alpha_M}$ converges; with bounded $M(\cdot, p)$ this yields a finite governor.

5. PETC Lawfulness (Prime-Exact Tensor Consistency)

We attach to each state a **prime signature** $\sigma(T) \in \mathbb{Z}^{(\mathcal{P})}$ (finitely supported) and a **multiplicity functional** $\mathfrak{M} : V \rightarrow \mathbb{Z}$. The runtime is **lawful** if the following invariants hold for all permitted primitives (linear combination, application of B_p , scaling, projection):

1. **Additivity:** $\sigma(T + S) = \sigma(T) + \sigma(S)$.
2. **Channel action:** For each p there exists an integer effect c_p s.t. $\sigma(B_p T) = \sigma(T) + c_p e_p$, where e_p is the unit signature at p .
3. **Conservation:** $\mathfrak{M}(T)$ is preserved by all primitives.
4. **Compatibility:** If an operation would violate 1–3, the runtime **fails closed**: it rejects the step and emits a minimally reproducible trace.

Theorem 5.1 (closure). If 1–3 hold for primitives, then they hold for any composition produced by the runtime. *Proof:* structural induction on program trees. $\square$

6. Acceptance Tests & Certificates (per step)

For each t , the runtime computes and logs:

- **Contraction margin:** $\gamma_t := 1 - X_t - \sum_p b_p |w_{p,t}| \geq \varepsilon_t$.
- **Projection KKT residual:** $r_t := \text{dist}(0, \partial \mathcal{L}(w_t, \lambda_t))$ (should be $\leq \tau$).
- **WAC/DSE check:** compute $\|K_{t+m-1} \cdots K_t\|$ on sliding windows and ensure $\leq \rho$ (or DSE analog).
- **Governor telemetry:** $\log \Xi_t$, $\sum_p M(T_t, p) p^{-\alpha_M}$, and correlation with budget utilization.
- **PETC ledger:** differences $\Delta \sigma, \Delta \mathfrak{M}$ (must be zero unless explicitly allowed by rule 2).

Passing all tests for all t yields a **step-wise certificate**; aggregating them over a horizon yields a **run certificate**.

7. Sufficient Conditions for Feasibility & Summability

- **(Finite support)** If $|\mathcal{P}_t| < \infty$ then feasibility is purely budgetary: choose $w_t \in \mathcal{S}_t$.
- **(Dirichlet-decay design)** Suppose $b_p \leq C p^\beta$ for some $\beta \geq 0$ and proposals obey $|\hat{w}_{p,t}| \leq A_t p^{-\alpha_P}$ with $\alpha_P > \beta + 1$. Then $\sum_p b_p |\hat{w}_{p,t}| \leq C A_t \sum_p p^{\beta - \alpha_P} < \infty$. Scaling by Ξ_t and projecting preserves feasibility.

8. Reference Algorithm (Projection-First Loop)

```

Input:  $F$ , bounds  $\{b_p\}$ , gap schedule  $\{\varepsilon_t\}$ , magnitude  $\{X_t\}$ , estimators  $M(\cdot, p)$ ,
guards  $\delta_{\min}$ 
State:  $T_0 \in V$ 
for  $t = 0, 1, 2, \dots$  do
    // Proposals from learner
    obtain active set  $\mathcal{P}_t$  and proposal weights
         $\widehat{w}_t = (\widehat{w}_{\{p,t\}})_{p \in \mathcal{P}_t}$ 

    // Governor scaling
     $\xi_t \leftarrow (\sum_{p \in \mathcal{P}_t} M(T_t, p) p^{-\alpha_M} + \delta_{\min})^{-1}$ 
     $\widetilde{w}_t \leftarrow \xi_t \cdot \widehat{w}_t$ 

    // ACE projection
     $R_t \leftarrow 1 - \varepsilon_t - X_t$ 
     $w_t \leftarrow \operatorname{argmin}_{\{w \in \mathbb{R}^{\mathcal{P}_t}\}} \frac{1}{2} \|w - \widetilde{w}_t\|_2^2 \quad \text{s.t.} \quad \sum_p b_p |w_p| \leq R_t$ 

    // Operator and update
     $K_t \leftarrow \sum_{p \in \mathcal{P}_t} w_{\{p,t\}} B_p$ 
     $T_{\{t+1\}} \leftarrow F + K_t T_t$ 

    // Certificates
    log  $y_t$ , KKT residual, WAC/DSE checks,  $\xi_t$  telemetry, PETC ledger
end for

```

9. Worked Micro-Example ($\mathbb{R}^2$)

Let $V = \mathbb{R}^2$ with Euclidean norm. Choose two channels with

$$B_2 = \begin{pmatrix} 0.2 & 0 \\ 0.1 & 0.3 \end{pmatrix}, \quad b_2 = 0.4; \quad B_3 = \begin{pmatrix} 0 & 0.3 \\ 0.2 & 0 \end{pmatrix}, \quad b_3 = 0.3.$$

Let $F = (1, 0)^\top$, $X_t = 0.1$, $\varepsilon_t = 0.1 \Rightarrow R_t = 0.8$. Suppose the learner proposes $\widehat{w}_{2,t} = 1.1$, $\widehat{w}_{3,t} = 1.0$ and $M(T_t, 2) = M(T_t, 3) = 1$ with $\alpha_M = 2$, $\delta_{\min} = 0$. Then $\Xi_t = (1 + 1/3 + \dots)^{-1} \approx 1/1.5 \approx 0.666$. After scaling, the ACE projection enforces

$$0.4|w_{2,t}| + 0.3|w_{3,t}| \leq 0.8 \Rightarrow (|w_{2,t}|, |w_{3,t}|) \approx (1.33, 0.89) \text{ (one admissible solution),}$$

so $\|K_t\| \leq 0.8$ and the step is contractive. The trajectory converges geometrically by Theorem 3.2.

10. Practical Checklists

- **Choose gaps:** pick $\varepsilon_t \in [\varepsilon_{\min}, \varepsilon_{\max}]$ with $\varepsilon_{\min} > 0$. Larger gaps give stronger certificates but tighter budgets.
 - **Track margins:** ensure $\gamma_t \geq \varepsilon_t$ and alert if drift threatens $R_t \rightarrow 0$.
 - **Window configuration:** set m for WAC and M for DSE; log the worst m - and M -window products.
 - **PETC:** validate signatures and multiplicities at primitive level; reject on first violation.
 - **Numerics:** implement weighted soft-threshold with sort-by $|\tilde{w}_p|/b_p$; emit KKT residuals.
-

11. Summary of Guarantees

1. **Per-step contraction** via ACE implies stability of the update map.
 2. **Global convergence** in the time-invariant case; **Cauchy convergence** (and fixed-point convergence if $K_t \rightarrow K_\infty$) under uniform contraction.
 3. **Windowed guarantees** under WAC/DSE enabling controlled time variation and occasional expansions.
 4. **Scalar governor** provides a tunable small-gain mechanism compatible with ACE.
 5. **Lawfulness** is preserved by construction via PETC invariants and fail-closed semantics.
-

12. Open Extensions

- Tight non-asymptotic bounds for WAC/DSE with state-dependent K_t .
 - Adaptive selection of α_M to optimize sensitivity vs. robustness.
 - Probabilistic certificates when proposals are stochastic.
 - Connections to spectral certification (gap tracking in the spectrum of K_t).
-

Version. v0.1 (this note).